Setting the window by posterior entropy

The criterion picks the right direction and the wrong operating point: it beats every adaptive variant tried and still loses to a fixed k = U1.

The criterion. Take the largest window whose posterior entropy is still above a floor h_min. Entropy falls with both k and time, so k shrinks along a generation: coarse scales while the grid is ambiguous, fine scales last.

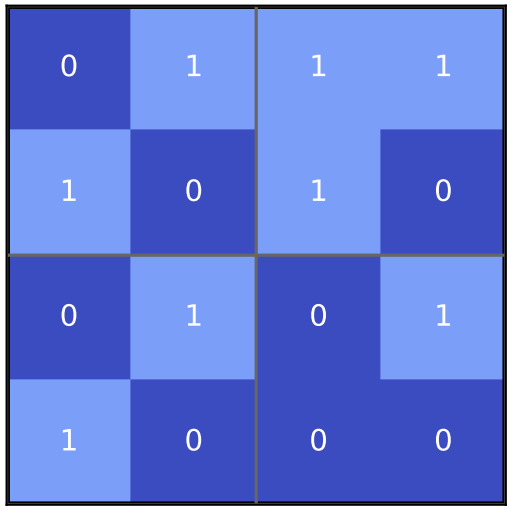
Scored for the first time on novel AND grammatical, not grammaticality alone. A machine that copies its training set scores 1.0 on grammaticality.

Re-read of the 2026-11-03 sweep: adaptive loses to the best fixed k in 20/21 cells (median success ratio 0.672) -- but beats the fixed k = 5 it actually imitates in 11/12. The choice set was wrong, not the adaptivity.

One variable changed (51098252): drop k = 1, leaving k >= U1 = 3. Now a binding floor wins in 15/15 cells where before it lost in 21/21, and the new machine beats the old in 15/15. It still loses to fixed k = 3 in 12/15.

Tokenization: what a "sample" is, and what the window sees

one sample: 4x4 grid of tokens in {0..1}



Tokens. Every site holds one of v = 2 symbols. There is no pixel intensity and no ordering: token 3 is not "between" 2 and 4. A grid is 4x4 = 16 categorical sites.

Grammar. A root symbol expands 2 times. Each symbol is replaced by one of m = 3 synonymous 2x2 "words", drawn once and then fixed. So the grey lines at left bound aligned 2x2 blocks (one level-3 symbol each) and the black lines bound aligned 4x4 blocks (one level-2 symbol each). A grid is grammatical iff some root derives it.

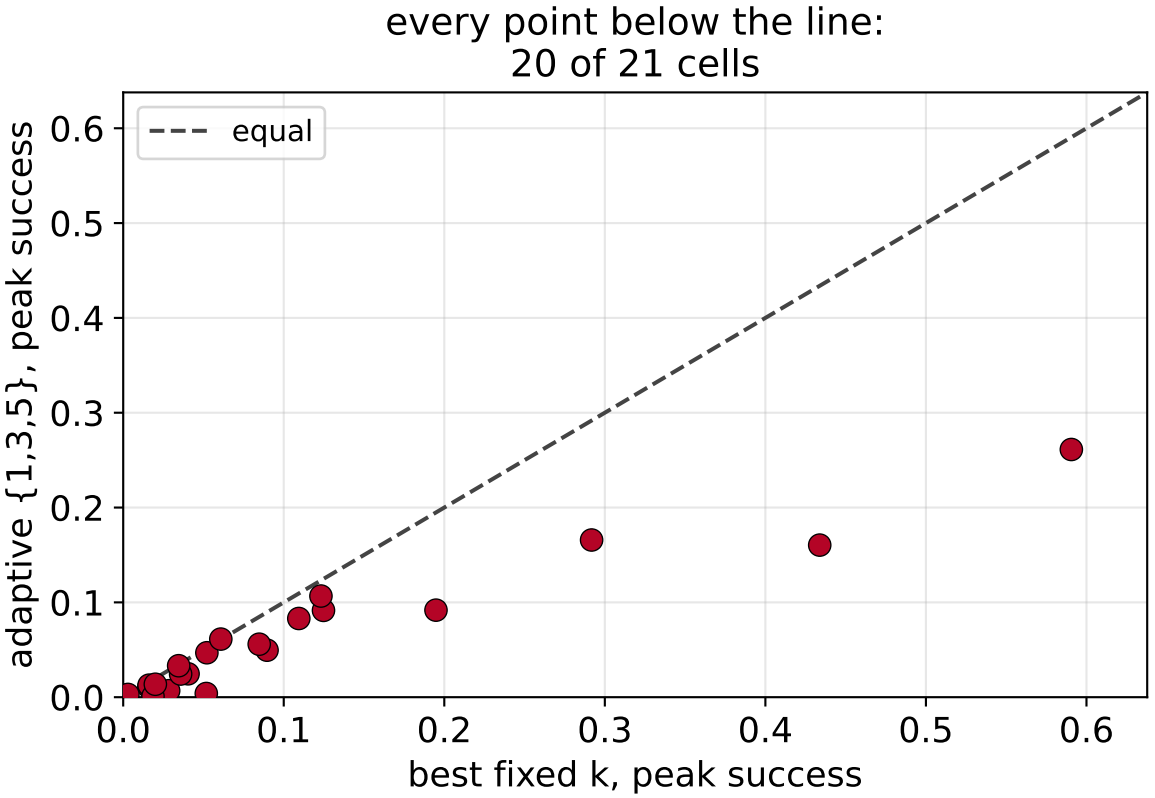
Diffusion. Absorbing-state ("erasure"): each site is independently replaced by a MASK token with probability p(t). Unmasked sites are never corrupted, so exact Bayes denoising is patch matching -- no score approximation enters.

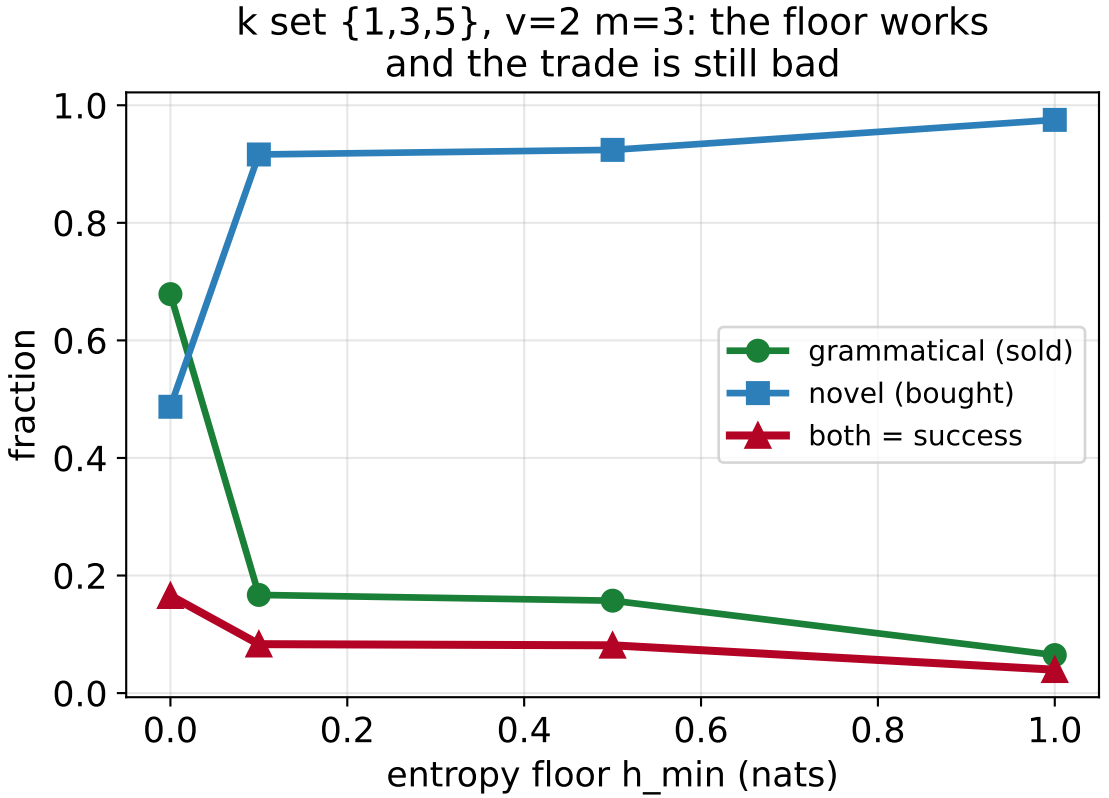
LS machine. The posterior at a masked site is the empirical distribution of the centre tokens of all training k x k patches that agree with the observed sites inside that window, with Laplace smoothing alpha. Sites are revealed one at a time (exact ancestral sampling; revealing in parallel from marginals adds ~n^2/2T independence errors, which alone made 12% of grids ungrammatical).

Success (hshunt's criterion). A generated grid counts only if it is NOT in the training set AND parses completely. Since every training grid is grammatical,

$$\text{success} = \text{grammatical} + \text{novel} - 1$$

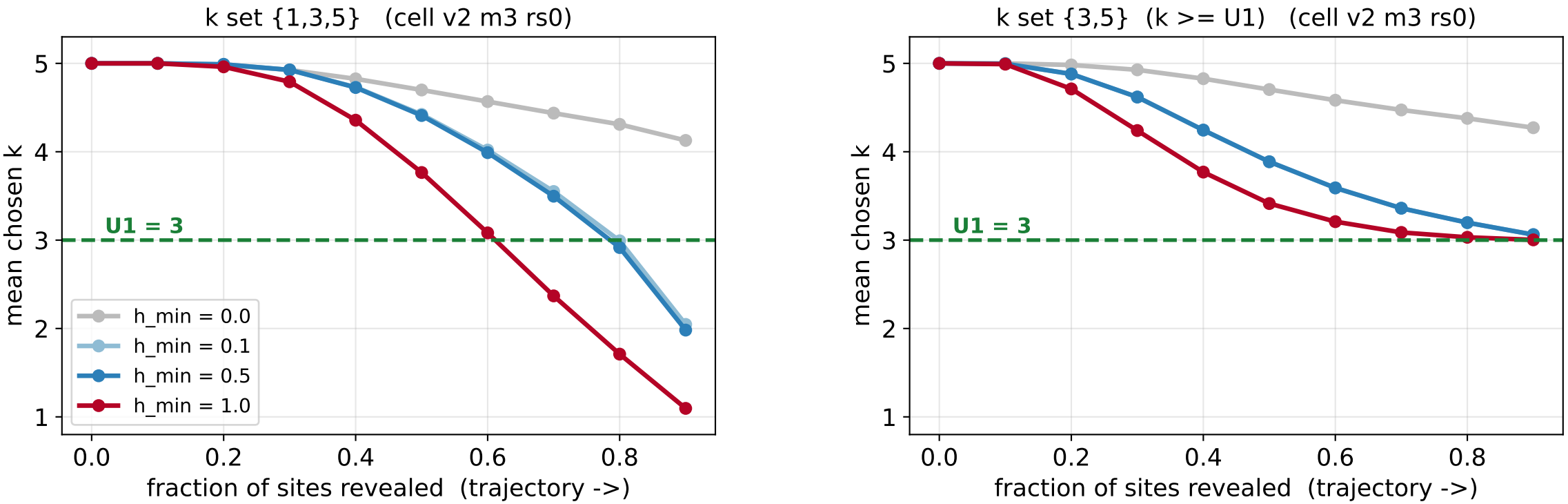
The re-read: grammaticality said yes, the real criterion said no





Left. Peak success over the shared training-set ladder, adaptive against the best fixed k on the same grammar and seeds. The adaptive machine is below the diagonal in 20 of 21 cells, median success ratio 0.672. It is not a close call and it is not noise: the same ordering holds in every grammar family measured (v = 2, 3, 4; m = 2, 3, 5; L = 2 and 3).
Right. Why. Raising the floor does exactly what it was designed to do -- novelty climbs 0.487 -> 0.975 as the mean window falls 4.67 -> 3.52 -- and grammaticality falls 0.679 -> 0.065 faster, so success drops 0.166 -> 0.040. The floor stops the machine memorizing and bankrupts it doing so. Its best setting was h_min = 0, i.e. no floor at all, in 21 of 21 cells.

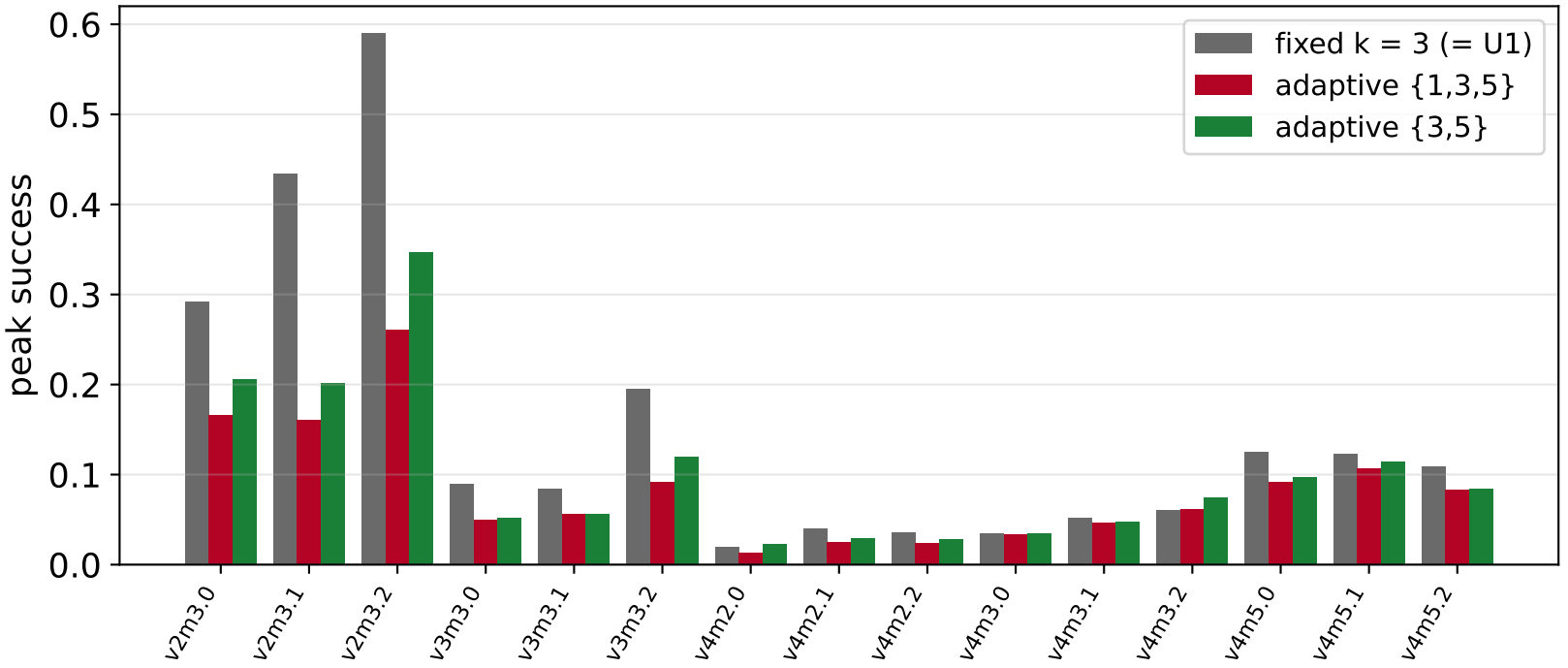
The induced k(t): the direction is right at every floor

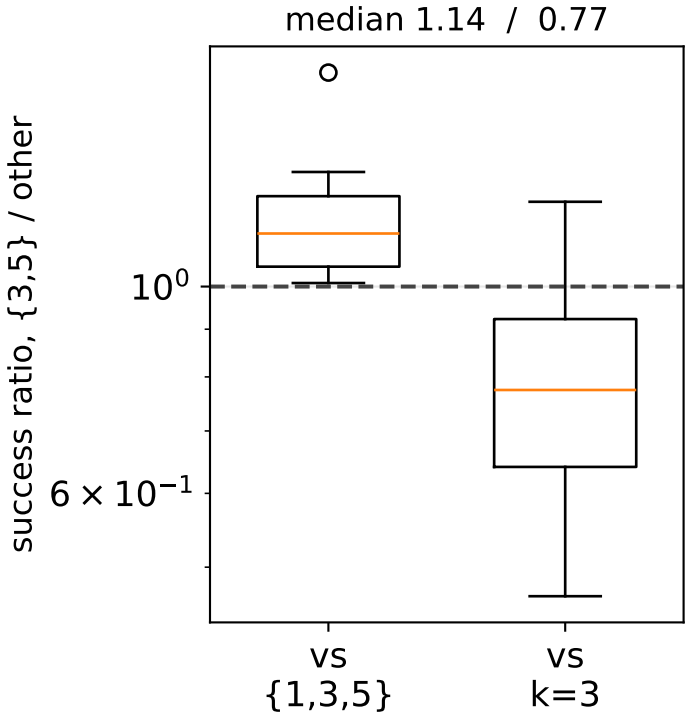


k shrinks monotonically along every trajectory at every floor, in both machines -- the direction hshunt predicted, and it is produced by the entropy criterion rather than imposed. Both panels are one cell, v2 m3 rs0, so the curves can be read; the win-counts on the other pages are over all 15-21 cells. With no floor the schedule is nearly flat here (5.00 at the first reveal, 4.13 at the last): the largest window almost always still matches something, so "largest that matched" barely varies. The floor is what gives the schedule range.

Left, the failure. At h_min = 1.0 the unrestricted machine ends the trajectory at mean k = 1.10, well below U1 = 2 s^(L-1) - 1 = 3, the smallest window that can contain a complete top-level child block. Right, the fix: with k = 1 removed the same floor lands at 3.00 and the collapse does not happen. Note the honest limit -- above U1 at L = 2 there are only two rungs, 3 and 5, so this is a thin test of a schedule, not a rich one.

Three machines, same grammars, same ladder, same seeds



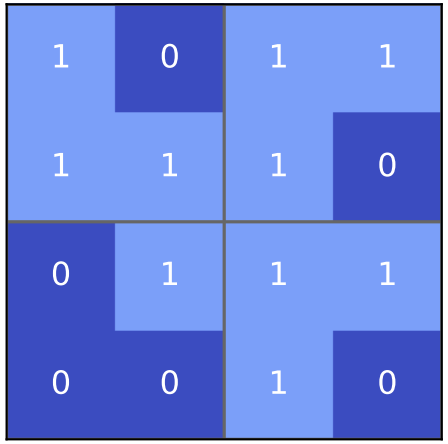


Removing the sub-U1 windows helps, unambiguously: the new machine beats the old in 15 of 15 cells (median ratio 1.14), and -- the real reversal -- its best floor is a binding one (h_min > 0) in 15 of 15 cells, where in the unrestricted machine h_min = 0 won in all 21. Once the floor can no longer select a window that cannot see the root, the entropy criterion starts paying.

It still is not enough. Fixed k = 3 wins in 12 of 15 cells, median ratio 0.77; the three wins are +0.0032, +0.0004 and +0.0142, which is not a result. The prediction registered before the run was that {3,5} with a binding floor would beat both the control and fixed k = 3. Half of that held. The null stated alongside it -- that at L = 2, U1 already is the schedule -- is the better fit.

Two generations, reveal by reveal: what the floor changes

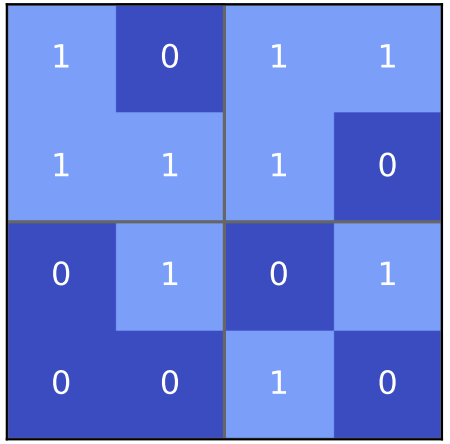
floor (h_min = 0.1)
novel: yes
grammatical: yes



step	site	k	H(nats)	token
(first 6 reveals omitted, k = 5 in both)				
6	9	5	0.562	1
7	2	5	0.562	1
8	12	5	0.501	0
9	7	5	0.637	0
10	5	3	0.562	1
11	13	3	0.002	0
12	0	5	0.003	1
13	10	3	0.693	1
14	6	3	0.693	1
15	15	3	0.002	0

mean k 4.38 total H 6.91 nats

control (h_min = 0.0)
novel: NO (copy)
grammatical: yes



step	site	k	H(nats)	token
(first 6 reveals omitted, k = 5 in both)				
6	9	5	0.562	1
7	2	5	0.562	1
8	12	5	0.501	0
9	7	5	0.637	0
10	5	5	0.008	1
11	13	5	0.008	0
12	0	5	0.003	1
13	10	5	0.008	0
14	6	5	0.008	1
15	15	5	0.004	0

mean k 5.00 total H 4.99 nats

Both columns are generation #27 of the same batch: same grammar and training set (v = 2, s = 2, L = 2, m = 3, P = 30, rule seed 0), same sampler, same random seed. Only h_min differs, so every difference below is caused by the floor. Sites are listed in the order the sampler revealed them, one per step, over all 16 sites. k is the window the entropy criterion chose at that site at that step, H the entropy of the posterior it was then sampled from.

The floor run holds a mean window of 4.38 against 5.00 for the control, and the gap opens late: both start at k = 5 while the grid is empty, and only the floored run steps down as the window fills and the posterior sharpens. Over these 64 generations the floor gives novelty 0.875 against 0.406, grammaticality 0.312 against 0.750, and success 0.188 against 0.156 -- the same direction as the 4096-generation sweep, which is the check that these two columns are typical.

What this says about setting the patch size

1. The direction is confirmed and it is not an assumption.

The entropy criterion makes k shrink along the generation at every floor and in both machines, because the observed sites inside a window only accumulate and the posterior only sharpens. Nobody imposed that schedule; it falls out of the floor.

2. The floor does stop memorization, exactly as intended.

On the unrestricted machine, h_min 0 -> 1.0 lifts novelty 0.460 -> 0.997, median over the 21 cells (0.487 -> 0.975 on the single cell plotted earlier). A working anti-copying device -- just not, by itself, a working sampler.

3. The failure was the choice set, and fixing it reversed the floor verdict.

Letting the floor pick k = 1 spends grammaticality on windows too small to contain a top-level child block. With k >= U1 the best floor becomes a binding one in 15/15 cells, against 0/21 before, and the machine beats its predecessor in 15/15.

4. It still loses to simply fixing k = U1 (12/15 cells, median ratio 0.775).

So at L = 2 the honest answer is the null: U1 already is the schedule. The criterion earns its keep against other adaptive machines, not against the fixed window.

5. The test is thin where it matters, and that is the next experiment.

Above U1 at L = 2 there are only two rungs, 3 and 5. A schedule has nowhere to go. L = 3 has U1 = 7 and real room, but its k = 7 tables overflow the int64 window code -- fixing that is the prerequisite for a fair test, and it is now the top of the queue.

Caveat on "cannot be grammatical". A window below U1 cannot see the root in one look, but overlapping windows still chain constraints: in 1D at n = 16 a k = 3 window scores 0.06 grammaticality, and the success optimum there sits at k = 11, below U1 = 15. "k >= U1" is the right floor for this 2D L = 2 machine, not a general law.